Saturated Superfluid Vacuum V
Condensate Black Holes

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Draft

Abstract

This paper develops the ontology of black holes in the Saturated Superfluid Vacuum (SSV) framework, where the vacuum is a real medium described by a macroscopic order parameter Ψ, matter consists of topological defects in Ψ, and gravity arises from the interference cross-term among those defects (Paper IV). The question is what happens at the extreme end of gravitational collapse.

The argument has four steps. Because matter is made of vortex defects with finite core size ξ, collapse terminates at the close-packing density of those cores rather than at an infinite-density singularity. The surface at which outward acoustic progress relative to the exterior vanishes — the acoustic-freezing surface, where the update availability $A(r) = 0$ — is identified as the horizon: a real physical surface of the medium, not a coordinate artifact. Spin is topological rather than material rotation, since $\oint \nabla\theta \cdot d\mathbf{l} = 2\pi n$ is a homotopy invariant that acoustic freezing does not erase, so angular momentum is conserved as winding. And because there is no singularity, information is not destroyed: infalling structure modifies the condensate state and its boundary, redistributing the information into present-state wake complexity in the sense of Paper III.

A fifth result connects to Paper III's update-capacity language. The acoustic horizon has a definite surface gravity κ, and a stationary observer near it must accelerate to remain in place; by the Unruh effect such an observer registers thermal radiation at $T \propto \hbar c^3/(GMk_B)$, recovering the *form* of the Hawking temperature. The numerical coefficient carries a factor-of-four gap of the same origin as the light-deflection gap in Paper IV — the leading-order Newtonian potential misses the spatial-curvature contribution — which Paper VII-b addresses.

The paper claims the ontology and the mechanism. It does not claim a full Kerr derivation, an exact Bekenstein–Hawking coefficient, a black-hole population model, or galaxy-scale or cosmological applications.

Contents

1 Introduction

This paper treats black holes in SSV as coherent condensate objects with finite interiors, acoustic-freezing horizons, topological spin, and information-storing boundary structure.

The purpose is deliberately narrow. The question is ontological: what is a black hole if the vacuum is a saturated superfluid medium rather than an empty spacetime background?

The SSV answer, developed in the following sections, is that gravitational collapse does not end at an infinite-density singularity. It ends at a finite-density coherent state of the vacuum medium, bounded by a surface at which outward acoustic progress tends to zero. This surface—the acoustic-freezing horizon—is not a coordinate artifact but a real physical feature of the medium. Spin is preserved as topological winding of the condensate phase. Information is not annihilated; it is written into the present-state complexity of the condensate and its boundary.

Dependencies

This paper draws on Papers I–IV in the SSV series. From Paper I [1]: the LogSE field equation, the vortex-defect ontology, and the healing length ξ. From Paper II [2]: the gravitational potential Φ as an interference cross-term and the fixed value of G. From Paper III [3]: proper time as the local change-rate of Ψ, and entropy as wake-complexity. From Paper IV [4]: the update availability $A = d\tau/dt = 1 + \Phi/c^2$ and the derivation of gravitational time dilation, deflection, and redshift from the one potential Φ.

What this paper does not claim

- No full Kerr metric derivation. Angular momentum in SSV is identified as topological winding; the exterior metric at all orders is deferred to Paper VII-b [5].
- No exact Bekenstein–Hawking coefficient. The area entropy is derived in form; the numerical prefactor requires the spatial-curvature correction and the identification $\xi = l_P$, both deferred.
- No black-hole population model, no claims about astrophysical near-extremal spin clustering.
- No galaxy-scale standing-wave applications and no broad cosmology.

Roadmap

Section 2 fixes the minimal SSV background needed for strong-gravity analysis. Section 3 argues that collapse terminates at finite density. Section 4 derives the horizon as an acoustic-freezing surface. Section 5 establishes spin as topological winding. Section 6 connects horizon structure to the wake-entropy framework of Paper III. Section 7 derives the Hawking temperature as an acoustic Unruh effect. Section 8 collects open problems.

2 SSV Setting for Strong Gravity

Only the parts of the SSV framework required for strong-gravity analysis are recalled here. The full derivations are in Papers I–IV.

The vacuum order parameter is

$$\Psi(\mathbf{x},t) = \sqrt{\rho(\mathbf{x},t)}\, e^{i\theta(\mathbf{x},t)}, \tag{1}$$

with density ρ, phase θ, and superfluid velocity $\mathbf{v}_s = (\hbar/m)\nabla\theta$ away from vortex cores. The field equation is the logarithmic Schrödinger equation (LogSE) [1]

$$i\hbar\,\partial_t\Psi = \left[-\frac{\hbar^2}{2m}\nabla^2 + b\hbar\ln\left(\frac{|\Psi|^2}{\rho_0}\right)\right]\Psi, \tag{2}$$

with coupling constant b and background density ρ_0. The Madelung decomposition of (2) gives a continuity equation

$$\partial_t\rho + \nabla\cdot(\rho\,\mathbf{v}_s) = 0, \tag{3}$$

and a Bernoulli equation whose nonlinear potential $b\hbar\ln(\rho/\rho_0)$ sets the bulk sound speed

$$c_s = \sqrt{\frac{b\hbar}{m}} \tag{4}$$

Notation. The symbol b here is a *frequency*, so that $b\hbar$ is an energy and $b\hbar/m$ a velocity squared. This is *not* Paper I's b, which is defined through $b\rho_0 = m_0c^2$. The two differ by a factor of $\hbar/\rho_0$ and the change of meaning was previously undeclared (#187). and the healing length

$$\xi = \frac{\hbar}{mc_s} = \sqrt{\frac{\hbar}{mb}}. \tag{5}$$

2.1 Update capacity and the gravitational potential

Paper IV introduces the update availability $A(\mathbf{x},t)$, defined as the local rate of proper-time advance per unit coordinate time,

$$A = \frac{d\tau}{dt} = 1 + \frac{\Phi}{c^2}, \tag{6}$$

where Φ is the gravitational potential derived from the interference cross-term among embedded defects. For a spherically symmetric mass M at $r \gg \xi$,

$$\Phi(r) = -\frac{GM}{r} \tag{7}$$

to leading order [2, 4]. Substituting into (6),

$$A(r) = 1 - \frac{GM}{rc^2}. \tag{8}$$

This expression has the Schwarzschild structure: A decreases from unity at $r \to \infty$ to zero at a finite radius, and turns negative for smaller r.

2.2 Outward acoustic propagation

An acoustic mode emitted radially outward from position r has, to leading order in Φ/c^2, a coordinate velocity

$$v_{\text{out}}(r) = c_s \cdot A(r). \tag{9}$$

For long-wavelength excitations of the condensate, $c_s \to c$ (the Bogoliubov modes propagate at c in the long-wavelength limit [1]), so (9) becomes

$$v_{\text{out}}(r) = c \cdot A(r). \tag{10}$$

The sign and magnitude of v_{out} track A: positive for $r > r_H$, zero at $r = r_H$, and negative inside. The acoustic horizon is therefore the surface $A(r_H) = 0$.

3 Gravitational Collapse Without Singularity

The Penrose singularity theorem [6] establishes, within general relativity, that gravitational collapse of ordinary matter must terminate in a geodesic-incompleteness singularity once a trapped surface has formed. SSV departs from this conclusion not by evading the theorem but by operating outside its domain. The medium described by the LogSE is not a classical or quantum-field-theory matter source embedded in a fixed spacetime; it is the spacetime-generating medium itself. The singularity theorems do not apply.

3.1 Close-packing of vortex cores

In SSV, all matter consists of topological vortex defects in Ψ. Each defect has a core of radius $\sim \xi$ at which $|\Psi| \to 0$ with nonzero phase winding. Two defects of the same winding sign repel when their cores approach within $\sim \xi$: the energy cost of overlapping density minima diverges as the cores overlap, because the phase-gradient kinetic energy grows as the inter-core separation shrinks below ξ.

The maximum defect density is therefore set by close packing of cores,

$$\rho_{\max} \sim \frac{E_{\text{core}}}{\xi^3}, \tag{11}$$

where $E_{\text{core}} \sim \hbar c_s/\xi$ is the core energy per unit length times ξ. Once the medium reaches $\rho_{\max}$, further compression would require overlapping cores, which is topologically forbidden. Collapse therefore terminates at finite density.

Gap (Paper I program): The exact value of $\rho_{\max}$, the threshold density at which repulsion arrests collapse, requires a full 3D static minimisation of the LogSE with defect close-packing boundary conditions. This is the open computation of the Paper I programme [1]. The qualitative claim—finite termination density—does not depend on that number.

3.2 The condensate endpoint

The endpoint of collapse is a finite-density coherent condensate state of the medium. Its defining properties are:

1. Interior density ρ_{int} finite and $\lesssim \rho_{\max}$.
2. Phase $\theta(\mathbf{x})$ and density $\rho(\mathbf{x})$ well-defined throughout the interior.
3. A causal boundary—the acoustic-freezing horizon—separating interior from exterior.

The condensate picture resolves the apparent paradox of supermassive black holes. A black hole of mass $M = 10^9\,M_\odot$ has Schwarzschild radius $r_S = 2GM/c^2 \approx 3 \times 10^{12}$ m and average density

$$\bar{\rho} = \frac{3M}{4\pi r_S^3} \approx 10^{-5}\,\mathrm{kg\,m^{-3}}, \tag{12}$$

far below nuclear density. There is no requirement that the condensate be locally Planck-dense throughout its volume; the defining feature is not local density but the formation of a horizon-scale acoustic-freezing surface.

4 Horizon as Acoustic-Freezing Surface

4.1 The freezing condition

Setting $A(r_H) = 0$ in (8):

$$r_H = \frac{GM}{c^2}. \tag{13}$$

At $r = r_H$, every outward acoustic mode has $v_{\mathrm{out}} = 0$: the acoustic horizon. For $r < r_H$, $A < 0$, so $v_{\mathrm{out}} < 0$: even nominally outward-directed modes are dragged inward in coordinate time. The region $r < r_H$ is causally disconnected from the exterior.

Gap (Paper VII-b): The Newtonian potential (7) misses the spatial curvature contribution to Φ, which is the same factor deferred in Paper IV §6. The full GR result is $r_S = 2GM/c^2$, a factor of two larger than (13). The mechanism and qualitative structure are unaffected; the exact coefficient is established once Paper VII-b [5] derives the spatial half of the metric.

4.2 Freezing versus pinning

The acoustic-freezing condition has a specific meaning. It does not assert that the condensate interior is spatially immobile in any absolute sense. It asserts that *outward signal propagation relative to the exterior observer* tends to zero.

The distinction is essential:

- *Freezing*: the failure of outward acoustic propagation, measured in coordinate time. This is what the horizon imposes.
- *Pinning*: absolute spatial immobility. This is not claimed.

A frozen condensate can still translate as a whole (its centre of mass can move), can carry angular momentum (Section 5), and can store information at its boundary (Section 6). These are global and topological properties that are not ordinary outward sound pulses escaping from the interior.

4.3 Coordinate time and proper time at the horizon

From Paper III, proper time is the local change-rate of Ψ. At $r = r_H$, $A = 0$: the interior clock advances at zero rate relative to the exterior coordinate. In the language of Paper III, the interior update-rate $\mathcal{R}[\Psi]$, as observed from outside, is zero.

Interior evolution does continue in the condensate's own local time. The condensate medium has definite Ψ at every interior point; it is not static in any intrinsic sense. The freezing is a causal isolation, not a cessation of interior change.

4.4 Summary statement

The claim of this section is therefore:

> The black-hole horizon is the surface at which the medium's update availability $A(r)$ vanishes. Outward acoustic progress freezes at this surface, causally isolating the interior condensate from the exterior. The surface is real and physical: it is a topographic feature of the update-capacity field $A(\mathbf{x})$, not a choice of coordinates.

5 Spin Without Acoustic Time

The most immediate objection to the condensate picture is: if time freezes at the horizon, why does a black hole spin?

The answer is that spin in SSV is not modeled as ordinary internal material rotation requiring freely propagating acoustic time through the interior. It is a topological property of the condensate phase field.

5.1 Circulation and winding number

The superfluid velocity is $\mathbf{v}_s = (\hbar/m)\nabla\theta$. The circulation around a closed loop $\mathcal{C}$ is

$$\Gamma_{\mathcal{C}} = \oint_{\mathcal{C}} \mathbf{v}_s \cdot d\mathbf{l} = \frac{\hbar}{m}\oint_{\mathcal{C}} \nabla\theta \cdot d\mathbf{l} = \frac{2\pi\hbar}{m}\, n_{\mathcal{C}}, \tag{14}$$

where $n_{\mathcal{C}} \in \mathbb{Z}$ is the winding number of the phase around $\mathcal{C}$. This is quantised and topologically protected: $n_{\mathcal{C}}$ is a homotopy invariant that changes only when a vortex line crosses $\mathcal{C}$.

5.2 Angular momentum as topological winding

For a condensate with axial symmetry and winding number n around the symmetry axis, the angular momentum is

$$J = \int_V \rho\,(\mathbf{r} \times \mathbf{v}_s)\, d^3x = \frac{\hbar\, n}{m}\, M_{\text{cond}}, \tag{15}$$

where $M_{\text{cond}} = \int \rho\, d^3x$ is the condensate mass. The dimensionless spin parameter

$$a \equiv \frac{Jc}{GM^2} = \frac{\hbar n c^2}{G m M_{\text{cond}}^2} \cdot M_{\text{cond}} = \frac{\hbar n c^2}{G m M_{\text{cond}}} \tag{16}$$

is a topological label of the condensate state.

5.3 Topological protection through the horizon

The winding number $n_{\mathcal{C}}$ for any loop $\mathcal{C}$ threading the condensate can change only if a vortex line crosses $\mathcal{C}$. For a loop encircling the condensate at $r > r_H$, the relevant mechanism would be expulsion of a vortex filament through the acoustic-freezing surface.

A vortex filament crossing the acoustic horizon must carry energy from the interior to the exterior. But the horizon is precisely the surface from which no acoustic energy propagates outward. A filament terminating at the horizon or expelled through it requires the medium to move outward across r_H—which the acoustic-freezing condition forbids for ordinary dynamical modes.

Penrose process and superradiance: yes, by emergent Kerr structure. External-perturbation extraction is the more interesting case, and the answer is clean: the analogue Penrose process and the analogue superradiance both occur in SSV, by the same mechanism as in the standard GR Kerr exterior. The reason is structural. The SSV-emergent metric of a rotating condensate black hole is the Kerr metric (Paper VII-b [5], §6.3, problem 2; the Jacobson route gives the full nonlinear Einstein equations, and Kerr is the unique stationary axisymmetric vacuum solution). Standard GR processes that depend on the Kerr ergoregion, namely the Penrose process [7] and bosonic superradiance, therefore lift to the SSV-emergent geometry without modification at the leading order treated here.

What is distinct in SSV is the *identification* of the extracted quantities. In Penrose's process, the particle that returns to infinity carries more energy than the incoming probe; in SSV this energy is acoustic energy of the surrounding medium, and the angular momentum carried by the returning particle is condensate winding transferred from the horizon weave to the external defect. The horizon weave's topological winding decreases by the amount transferred, exactly as the Kerr horizon's angular-momentum parameter a decreases in the GR calculation. Total winding is still conserved globally because the external defect carries off the difference.

The topological-protection claim of the previous paragraph is compatible: it concerns *spontaneous* expulsion, where no external driving carries the winding away. Penrose extraction and superradiance require explicit external coupling to the ergoregion, and that coupling is what transfers the winding outward. The two statements describe different regimes and do not conflict.

5.4 Separation of two claims

Two claims that can be conflated should be kept separate:

- *Spin persistence:* a condensate black hole carries angular momentum as topological winding. This is the claim of the present paper.
- *Population claim:* astrophysical black holes cluster near particular spin values. This requires separate observational and dynamical support and is explicitly outside the scope of this paper.

6 Horizon Entropy and Information Storage

6.1 From wake complexity to horizon entropy

Paper III establishes that entropy in SSV is the irreducible wake-complexity of the present state of Ψ:

$$S_{\text{wake}} = k_B \, W[\Psi(t)], \tag{17}$$

where $W[\Psi]$ measures the amount of present-state structure required to encode the surviving wake of prior events. For an isolated black-hole condensate, the relevant present structure is the condensate interior plus its horizon boundary.

Infalling matter modifies the condensate state. From the exterior, this modification is written into the boundary structure of the condensate at $r \approx r_H$, because that is the last surface at which incoming information can leave a physical impression before being causally isolated. The information is not destroyed; it is redistributed into the wake complexity of the horizon-scale boundary.

6.2 Area entropy from condensate surface counting

The surface gravity argument of Section 7 will establish that the relevant physical scale at the horizon is the healing length ξ. The horizon surface of area A_H can be partitioned into cells of size ξ^2 (the smallest independent region of the condensate surface). The number of independent configurational states of the condensate boundary is, schematically,

$$\Omega_H \sim \exp\left(\frac{A_H}{\xi^2}\right). \tag{18}$$

From the Boltzmann entropy,

$$S_H = k_B \ln \Omega_H \sim k_B \frac{A_H}{\xi^2}. \tag{19}$$

The Bekenstein–Hawking area law [8, 9] is $S_{\rm BH} = k_B c^3 A_H/(4G\hbar) = k_B A_H/(4l_P^2)$, with $l_P = \sqrt{\hbar G/c^3}$ the Planck length. Equation (19) reproduces the area dependence. The coefficient requires $\xi = l_P$, which holds when the constituent boson mass equals the Planck mass:

$$\xi = l_P \iff \frac{\hbar}{mc_s} = \sqrt{\frac{\hbar G}{c^3}} \iff m = \sqrt{\frac{\hbar c}{G}} = m_P. \tag{20}$$

Gap (Paper I program): The identification $m = m_P$—and hence the exact coefficient in (19)—requires deriving the constituent boson mass of the fundamental SSV condensate from the LogSE. This is part of the open computation of the Paper I programme. The area dependence of S_H is established without it.

Gap (Paper VII-b): The horizon area A_H also carries the factor-of-four correction from the spatial-curvature contribution to the horizon radius: $A_H^{\rm GR} = 4\pi r_S^2 = 4\pi(2GM/c^2)^2$ versus $A_H^{\rm SSV} = 4\pi r_H^2 = 4\pi(GM/c^2)^2$. The exact Bekenstein–Hawking coefficient absorbs both this factor and (20); both are deferred to Papers I and VII-b.

The area dependence is derived, not counted (H-EOS S2, #155). Equations (18)–(19) above obtain the area law by a *schematic* count of independent boundary cells of size ξ^2. The holographic-compliance programme (Paper VII-b; the gravity-sector decision recorded in #154) requires the area law to come instead from the medium's *cross-horizon entanglement entropy*—the area-law entropy of a short-range-correlated ground state with a UV cutoff ξ (Srednicki [10]; Bombelli *et al.* [11]), the input Jacobson's entanglement-equilibrium route [12] turns into the field equation. The battery `horizon_entanglement_entropy` computes this entropy directly, by the correlator method (Peschel [13]; Casini–Huerta [14]) applied to the Bogoliubov vacuum of the saturated medium, and finds it *area-law*: the entanglement-entropy exponent tracks a provably area-law gapped-scalar calibrator and sits a full unit in exponent below the volume-law (extensive thermal) control, with $S/R^3 \to 0$; the coefficient η is size-independent (universal to 5%) and reproduces across a spherically-symmetric acoustic-metric inhomogeneity at the horizon scale. The one candidate non-locality—the chiral-shear term—is *exactly silent* at quadratic order over irrotational flow (#138), so the form is chiral-protected. This *earns* the area dependence of (19) from the medium's vacuum, replacing the boundary count.

What is *not* earned: the coefficient. The whole of the remaining gap of the gapboxes above is localised into the single number $\eta = S_{\mathrm{ent}}/\xi^2$: a natural $\eta \sim 1/\xi^2$ overshoots G_N by $(a_p/\ell_P)^2 = 1/\alpha_G = (m_P/m_p)^2 \approx 1.69 \times 10^{38}$ (the Susskind–Uglum species problem, Sakharov's induced gravity in another guise). G's magnitude is therefore conceded as a sub-grain (trans-Planckian) input (#154), not derived. Result note: `h-eos-s2-issue155`. This branch delivers the area-law *form* only; the surface gravity, Unruh temperature, and the Clausius closure are the analytic battery S1, deferred.

6.3 No information destruction

The condensate picture dissolves the information-loss problem rather than solving it. The problem assumes that collapse terminates at a singularity where physical structure is destroyed. In SSV, there is no such endpoint. The condensate retains a definite Ψ everywhere inside r_H. Infalling structure changes the condensate state; the exterior loses *access* to the fine interior organisation, but access loss is not physical annihilation.

This is consistent with the wake-entropy framework of Paper III. The distinction between "information inaccessible to the exterior" and "information destroyed" is the distinction between high coarse-grained entropy (the exterior observer cannot resolve the fine wake structure) and literal erasure of the structure from Ψ. The medium keeps its records; the observer simply cannot read them from outside.

7 Thermal Emission from the Acoustic Horizon

7.1 The acoustic Unruh effect

Unruh [15, 16] showed that an observer with proper acceleration a in flat spacetime registers thermal radiation at temperature

$$T_U = \frac{\hbar a}{2\pi k_B c}. \tag{21}$$

The same effect arises at any acoustic horizon in a medium where the flow velocity equals the local sound speed [16, 17]: a stationary observer just outside the sonic horizon must accelerate to maintain position against the flow, and registers thermal phonons at the temperature set by its acceleration.

In SSV, the acoustic-freezing surface at $r = r_H$ plays the exact role of the sonic horizon. An observer who remains stationary at r slightly above r_H must sustain a proper acceleration to resist the infall tendency of the medium. This acceleration is set by the gradient of the update availability.

7.2 Surface gravity and Hawking temperature

A stationary observer at radius r must accelerate with proper acceleration

$$a(r) = c^2 \left| \frac{dA}{dr} \right| = c^2 \frac{GM}{r^2 c^2} = \frac{GM}{r^2} \tag{22}$$

to remain at fixed r in the potential $\Phi = -GM/r$. This is the Newtonian gravitational acceleration, confirmed from the same $A(r)$ that yields the time dilation of Paper IV.

At the horizon $r = r_H = GM/c^2$:

$$\kappa \equiv c^2 \left| \frac{dA}{dr} \right|_{r_H} = c^2 \cdot \frac{GM}{r_H^2 c^2} = \frac{c^4}{GM}. \tag{23}$$

The surface gravity κ is the coordinate acceleration needed to hover at the horizon. Substituting into (21),

$$T_H^{\mathrm{SSV}} = \frac{\hbar\kappa}{2\pi k_B c} = \frac{\hbar c^3}{2\pi k_B G M}. \tag{24}$$

The standard Hawking temperature [18, 9] is

$$T_H^{\mathrm{GR}} = \frac{\hbar c^3}{8\pi k_B G M}. \tag{25}$$

The two expressions agree in their dependence on $\hbar$, c, G, M, and k_B. The numerical factor differs by four.

Gap (Paper VII-b): The factor-of-four discrepancy has the same origin as the light-deflection and horizon-location gaps: the leading-order potential $\Phi = -GM/r$ places the horizon at $r_H = GM/c^2$ instead of the GR value $r_S = 2GM/c^2$, and the GR surface gravity includes the redshift factor at the horizon. Both corrections are recovered once the spatial-curvature contribution to the metric is derived in Paper VII-b.

Surface gravity, Unruh T, and trans-Planckian robustness from the flowing medium (H-EOS S1, #155). The acoustic metric is *kinematic*—fixed algebraically by the flow with no back-reaction (Visser [19], §14: "form yes, G no"). A concrete mass-conserving radial-sink dumb-hole $v(r) = c_s(r_H/r)^2$ (a radial sink, not a pure vortex, is required for a true horizon) crosses the density-independent LogSE sound speed $c_s = \sqrt{B_0}$ at r_H, and the battery `dumb_hole_surface_gravity` reproduces the Visser analogue-gravity forms to a relative 10^{-6}:

$$g_H = \tfrac{1}{2}\big|\partial_r(c_s^2 - v^2)\big|_{r_H} = \frac{2c_s^2}{r_H}, \qquad k_B T_H = \frac{\hbar g_H}{2\pi c_s} \;\Rightarrow\; T_H = \frac{c_s}{\pi r_H}. \tag{26}$$

The robustness margin (the negative-capable test; W4 / #148). The LogSE phonon dispersion $\omega^2 = c_s^2 k^2 + k^4/4$ is *superluminal* for all $k > 0$ (group velocity $v_g = (c_s^2 + k^2/2)/\sqrt{c_s^2 + k^2/4} > c_s$, excess $\frac{3}{8}k^2/c_s$), so high-k modes can leak across the horizon (Corley–Jacobson [20] discusses the general sensitivity of Hawking spectra to modified dispersion, but its numerical model is subluminal and does not establish robustness for this opposite-sign LogSE branch). The dimensionless scale-separation parameter is

$$M \equiv \frac{g_H\,\xi}{2\pi c_s^2} = \frac{\xi}{\pi r_H} \ll 1.$$

It is $\sim 10^{-40}$ for astrophysical horizons but $O(1)$ for the *local Rindler horizons* of the Jacobson construction at grain-scale acceleration. Thus $M \ll 1$ is a necessary scale-separation diagnostic, not a proved sufficient condition for thermality. A LogSE-specific superluminal mode-conversion calculation is still required before robustness can be claimed.

The Jacobson route closes in form. With T_H here and the area-law $dS = \eta\, dA$ of §6 (S2), the Clausius relation $\delta Q = T_H\, dS$ gives $\nabla^2\Phi = 4\pi G_{\mathrm{eff}}\, e/c^2$ with $G_{\mathrm{eff}} = 1/(4\hbar c\,\eta)$. The measured η is $O(1)$ per ξ^2, so G_{eff} overshoots G_N by $(a_p/\ell_P)^2 \approx 1.69 \times 10^{38}$—the conceded sub-grain input (#154); the identity $(a_p/\ell_P)^2 = 1/\alpha_G$ is the guarded #122 tautology, not a derivation. Result note: `h-eos-s1-issue155`. *Form yes, G no*, through the full chain.

7.3 Qualitative consequences

Even without the numerical coefficient, the SSV derivation establishes three points.

Hawking radiation is not a quantum-gravity phenomenon in SSV. It is the acoustic Unruh effect of a condensate with a real physical horizon—the same effect that has been observed in analogue acoustic black holes in Bose–Einstein condensates [21]. No new physics beyond the LogSE and the update-capacity framework is needed.

The temperature is set by the horizon gradient. From (24), $T_H \propto c^4/(GM)$. Smaller (lighter) black holes have higher Hawking temperatures. As the condensate radiates, M decreases, r_H decreases, and T_H increases—the well-known Hawking evaporation runaway.

The endpoint of evaporation is not a naked singularity. As $r_H \to \xi$, the condensate reaches the Planck scale. At this point the close-packing regime of Section 3 is reached and the condensate cannot compress further. The evaporation terminates at a Planck-scale remnant rather than a singularity; the next subsection establishes that the remnant is structurally stable rather than dissolving back into the medium.

7.4 The Planck-scale evaporation endpoint

The qualitative termination above leaves a structural question open: does the Planck-scale remnant dissolve completely into the saturated medium, or does it stabilise as a topologically protected defect? The close-packing equation of state of §3 answers this definitively. The LogSE equation of state has three properties that together fix the endpoint as a *stable* Planck-scale remnant.

Argument 1: confinement of sub-saturated regions. On the adopted branch (Paper I, #183) the saturation potential is $V(\rho) = +b\rho[\ln(\rho/\rho_0) - 1] + V_0$, with chemical potential and pressure

$$\mu(\rho) = \frac{dV}{d\rho} = +b\ln(\rho/\rho_0), \qquad P(\rho) = \rho\mu - V = b\rho. \tag{27}$$

V is convex with its minimum at ρ_0 and is bounded below, so the uniform saturated state is the global minimum. Because P is *strictly increasing* in ρ, a sub-saturated region carries lower pressure than the medium around it and is therefore **compressed**: the surrounding medium exerts a net inward push $\Delta P = b(\rho_0 - \rho)$ on any region that tries to fall below saturation. A remnant attempting to "dissolve" must expand below saturation against this pressure.

There is no chemical-potential floor (#187). On the adopted branch $\mu \to -\infty$ as $\rho \to 0$, so confinement cannot be argued from a divergent μ. It follows from the pressure gradient instead, as set out above. Machine-checked in `ssv_v_remnant_audit_2026`.

Argument 2: topological-charge conservation. The original collapsing condensate carries integer winding number n around its axis (Eq. (15), §5). Hawking evaporation radiates phonons in the longitudinal pressure channel; these are pure phase disturbances of the saturated medium and carry zero winding. Throughout evaporation, n is therefore conserved. At the endpoint a fully dissolved ($n = 0$) state cannot be reached from a non-trivial parent ($n \neq 0$) without an unwinding event that passes through $\rho \to 0$ over an extended region. Argument 1 makes this costly: creating and expanding such a region works against the confining pressure $\Delta P = b(\rho_0 - \rho)$. Note the weakened form: with the chemical-potential floor withdrawn this is a finite *energy barrier*, not a prohibition, so what follows is suppression of unwinding rather than its impossibility (#187). Any black hole with $J \neq 0$ is therefore expected to leave a winding-bearing remnant, with the barrier height — and hence the tunnelling rate through it — an open computation.

Argument 3: horizon vanishing at the Planck scale. The acoustic-freezing mechanism of §4 requires a smooth $A(r)$ profile with $A \to 0$ over a region of scale $\gtrsim \xi$ for the continuum acoustic approximation to apply. When $r_H \lesssim \xi$ the smooth horizon can no longer be supported by the medium: the very mechanism that produces Hawking radiation switches off. The remnant ceases to be a "black hole" in the acoustic-horizon sense and becomes a static Planck-scale condensate defect, no longer radiating. This terminates evaporation at a definite mass scale.

Endpoint mass and structure. Combining the three arguments, the evaporation endpoint is a static Planck-scale defect with

$$M_{\rm rem} \sim \frac{c^2\,\xi}{G} \sim m_P, \qquad r_{\rm rem} \sim \xi \sim \ell_P, \tag{28}$$

carrying the parent black hole's topological winding n and confined by the pressure of the surrounding saturated medium (Argument 1 as corrected; the phrase "chemical-potential pressure" used previously named the withdrawn mechanism). The defect cannot annihilate spontaneously: doing so would require either dissolution (forbidden by Argument 1) or topological-charge violation (forbidden by Argument 2). Annihilation requires meeting an oppositely-wound partner, exactly as for any topologically protected defect in a superfluid.

Structural answer: stable Planck-scale remnant.

The close-packing equation of state of §3 forces the evaporation endpoint to be a *stable* Planck-scale defect, not a dissolved state. Three independent arguments converge:

1. the LogSE chemical potential diverges as $\rho \to 0$, so the saturated medium confines any sub-saturated region;
2. the topological winding n is conserved under Hawking evaporation (phonons carry no winding); a winding-bearing parent cannot evaporate to the trivial state without violating Argument 1;
3. the acoustic-horizon mechanism switches off at $r_H \sim \xi$, so the remnant ceases to radiate at the Planck mass.

Status: *structural*. The mass scale $M_{\rm rem} \sim m_P$ inherits the $\xi \sim \ell_P$ identification (Paper VII-b [5], candidate-grade); the no-dissolution conclusion is independent of the quantitative close-packing density $\rho_{\rm max}$ tracked under the Paper I programme [1]. The qualitative answer to the issue is therefore definite without the deferred numerics.

Observational consequence: Planck-mass relics. Every black hole that completes Hawking evaporation leaves behind a single Planck-mass topological defect. Under the standard Hawking evaporation estimate [9], primordial black holes (PBHs) with initial mass $M_* \sim 5 \times 10^{14}$ g would have finished evaporating in the present cosmological epoch, depositing $\sim M_*/m_P \sim 10^{19}$ times less remnant mass than parent mass — a remnant density today of $\Omega_{\rm rem} \sim 10^{-19}\,\Omega_{\rm PBH,initial}$, well below current dark-matter constraints unless the initial PBH abundance is near the present dark-matter density. Heavier PBHs ($M \gg M_*$) have not yet reached the endpoint and contribute negligibly to the relic density. The framework therefore predicts a definite but typically small remnant contribution to dark matter, with the size set by the PBH formation spectrum rather than by the endpoint mechanism itself.

8 Discussion and Open Problems

8.1 Correspondence with general relativity

The condensate picture reproduces the following features of GR black holes:

- A causal horizon that prevents outward signal propagation.
- Gravitational time dilation growing without bound as $r \to r_H$ (since $A \to 0$).
- Hawking radiation with the correct parametric dependence on $\hbar$, c, G, M.
- Angular momentum as a conserved parameter characterising the exterior field.
- Horizon entropy proportional to surface area.

The correspondence not claimed here:

- No Kerr metric derivation. The exterior field of a spinning condensate is characterised by M and J, but the exact spacetime geometry (including frame-dragging) requires the full emergent-metric treatment of Paper VII-b.
- No exact Bekenstein–Hawking coefficient. This requires $m = m_P$ and the spatial-curvature correction to r_H, both deferred.
- No PPN extension beyond leading order.

8.2 The no-hair structure

In GR, the no-hair theorem states that the exterior of a stationary black hole is completely characterised by mass M, angular momentum J, and electric charge Q. In SSV, the exterior acoustic field generated by the interference cross-term is determined by the same three quantities:

- M from the total condensate mass (integrated interference cross-term, Paper II).
- J from the topological winding number n, via (15).
- Q from a global $U(1)$ phase of Ψ, if the condensate carries charge.

Any other structure of the condensate interior would need to appear in the exterior field as a multipole moment above the monopole and dipole—but the LogSE cross-term in the far-field limit gives only a monopole (the Newtonian $1/r$ potential) and a dipole (the frame-dragging-like term from the axial circulation). Higher multipoles are exponentially suppressed at $r \gg r_H$.

Gap (Paper VII-b): A formal proof that the exterior field has only M, J, Q to all orders, and the derivation of the full Kerr-like exterior metric, are deferred to Paper VII-b.

8.3 Relation to analogue-gravity programme

The identification of the SSV horizon as an acoustic-freezing surface is not merely an analogy. In SSV, the acoustic horizon *is* the physical black-hole horizon. The condensate medium that produces gravity is the same medium whose excitations propagate at c and whose phonon spectrum terminates at the acoustic-freezing surface. The analogue-gravity programme [17] provides the mathematical tools for the acoustic Hawking effect; SSV provides the physical ontology that makes the correspondence exact rather than approximate.

8.4 Open problems

The following technical tasks remain for later work:

1. **Close-packing density:** Numerical determination of $\rho_{\max}$ from the 3D static minimisation of the LogSE with vortex close-packing boundary conditions (Paper I programme).

2. **Boson mass:** Derivation of $m = m_P$ (or the corresponding $\xi = l_P$) from the fundamental condensate specification (Paper I programme).

3. **Spatial curvature:** The factor-of-two correction to r_H and the factor-of-four correction to T_H and S_H, from the full emergent metric (Paper VII-b).

4. **Kerr exterior:** The exterior metric of a spinning condensate, including frame-dragging and the exact no-hair result (Paper VII-b).

5. **Evaporation endpoint:** The qualitative final state of Hawking evaporation as $r_H \to \xi$ is established as a stable Planck-scale topological remnant in §7.4, on three structural grounds (LogSE chemical-potential floor, topological winding conservation, acoustic-horizon vanishing at $r_H \sim \xi$). The quantitative close-packing density $\rho_{\max}$ and the precise relation $\xi = \ell_P$ remain in the Paper I programme and Paper VII-b respectively; neither affects the no-dissolution conclusion.

6. **Superradiance:** Whether angular momentum can be extracted from the condensate by external perturbations, and whether a Kerr-like superradiance instability exists.

9 Conclusion

In SSV, black holes are finite coherent condensates of the vacuum medium. Collapse terminates not at a singularity but at the close-packing density of topological vortex cores. The horizon is the acoustic-freezing surface $A(r_H) = 0$, a real topographic feature of the medium's update-capacity field. Spin is topological winding of the condensate phase, protected by homotopy against acoustic freezing. Information is written into the present-state wake complexity of the condensate and its boundary rather than destroyed. The horizon emits thermal phonons at the acoustic Unruh temperature $T_H \propto \hbar c^3/(GMk_B)$, recovering Hawking radiation as an ordinary consequence of the medium's acoustic structure.

Three gapboxes carry forward the same spatial-curvature correction deferred in Paper IV: the exact horizon radius, the Bekenstein–Hawking coefficient, and the Hawking temperature prefactor each require the emergent-metric treatment of Paper VII-b. Two further gapboxes are inside the Paper I numerical programme: the close-packing density and the identification $m = m_P$. Nothing in the present paper depends on those numbers for its qualitative claims.

References

[1] S. Norland, *Saturated Superfluid Vacuum I: Topological Defects in a Saturated Superfluid Vacuum — The Geometric Origin of Matter*, 10.5281/zenodo.19651986, 2026.

[2] Norland, S., "Saturated Superfluid Vacuum II: Forces as Hydrodynamic Pressure Gradients in the Saturated Superfluid Plenum," SSV Series Paper II (2026).

[3] Norland, S., "Saturated Superfluid Vacuum III: Thermodynamics and Irreversible Time," SSV Series Paper III (2026).

[4] Norland, S., "Saturated Superfluid Vacuum IV: Gravity as Update-Capacity Gradient," SSV Series Paper IV (2026).

[5] S. Norland, *Saturated Superfluid Vacuum VII-b: Emergent Geometry and the Dissolution of Quantum Gravity*, SSV programme, forthcoming.

[6] R. Penrose, "Gravitational collapse and space-time singularities," *Phys. Rev. Lett.* **14**, 57–59 (1965).

[7] R. Penrose, "Extraction of rotational energy from a black hole," *Nat. Phys. Sci.* **229**, 177–179 (1971).

[8] J. D. Bekenstein, "Black holes and entropy," *Phys. Rev. D* **7**, 2333–2346 (1973).

[9] S. W. Hawking, "Particle creation by black holes," *Commun. Math. Phys.* **43**, 199–220 (1975).

[10] M. Srednicki, "Entropy and area," *Phys. Rev. Lett.* **71**, 666–669 (1993).

[11] L. Bombelli, R. K. Koul, J. Lee, and R. D. Sorkin, "Quantum source of entropy for black holes," *Phys. Rev. D* **34**, 373–383 (1986).

[12] T. Jacobson, "Entanglement equilibrium and the Einstein equation," *Phys. Rev. Lett.* **116**, 201101 (2016).

[13] I. Peschel, "Calculation of reduced density matrices from correlation functions," *J. Phys. A* **36**, L205–L208 (2003).

[14] H. Casini and M. Huerta, "Entanglement entropy in free quantum field theory," *J. Phys. A* **42**, 504007 (2009).

[15] W. G. Unruh, "Notes on Black-Hole Evaporation," *Phys. Rev. D* **14**, 870–892 (1976).

[16] W. G. Unruh, "Experimental Black-Hole Evaporation?" *Phys. Rev. Lett.* **46**, 1351–1353 (1981).

[17] C. Barceló, S. Liberati, and M. Visser, "Analogue gravity," *Living Rev. Rel.* **8**, 12 (2005).

[18] S. W. Hawking, "Black hole explosions?," *Nature* **248**, 30–31 (1974).

[19] M. Visser, "Acoustic black holes: horizons, ergospheres, and Hawking radiation," *Class. Quantum Grav.* **15**, 1767–1791 (1998).

[20] S. Corley and T. Jacobson, “Hawking spectrum and high frequency dispersion,” *Phys. Rev. D* **54**, 1568–1586 (1996).

[21] J. Steinhauer, “Observation of quantum Hawking radiation and its entanglement in an analogue black hole,” *Nat. Phys.* **12**, 959–965 (2016).

A Code and Issue References

Each reference below is pinned so it can be checked directly against the repository (https://github.com/StigNorland/SVT): issues link to their tracker entry, and each script or report links to the exact version — the commit that last modified it. Issue numbers in the text hyperlink to this list.

Issues.

- #122 — https://github.com/StigNorland/SVT/issues/122
- #138 — https://github.com/StigNorland/SVT/issues/138
- #148 — https://github.com/StigNorland/SVT/issues/148
- #154 — https://github.com/StigNorland/SVT/issues/154
- #155 — https://github.com/StigNorland/SVT/issues/155
- #183 — https://github.com/StigNorland/SVT/issues/183
- #187 — https://github.com/StigNorland/SVT/issues/187

Code (each pinned to the commit that last modified it).

- `instruments/paper_v/dumb_hole_surface_gravity.py` @ f204600 — https://github.com/StigNorland/SVT/blob/f204600/instruments/paper_v/dumb_hole_surface_gravity.py
- `instruments/paper_v/horizon_entanglement_entropy.py` @ 9da99ea — https://github.com/StigNorland/SVT/blob/9da99ea/instruments/paper_v/horizon_entanglement_entropy.py
- `instruments/paper_v/ssv_v_remnant_audit_2026.py` @ ea8e776 — https://github.com/StigNorland/SVT/blob/ea8e776/instruments/paper_v/ssv_v_remnant_audit_2026.py

Reports (result notes, pinned to their last commit).

- `papers/SSV-V/results/h-eos-s1-issue155.md` @ f204600 — https://github.com/StigNorland/SVT/blob/f204600/papers/SSV-V/results/h-eos-s1-issue155.md
- `papers/SSV-V/results/h-eos-s2-issue155.md` @ 9da99ea — https://github.com/StigNorland/SVT/blob/9da99ea/papers/SSV-V/results/h-eos-s2-issue155.md

Change record. This paper states its *current* status only. Corrections, withdrawals and reframings are recorded in https://github.com/StigNorland/SVT/blob/main/papers/SSV-V/CHANGELOG.md, and the full history is in the repository's commits.